

COSIMA Cookbook: a computational learning module for analysing ocean and sea-ice model output

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

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Submitted: 01 January 1970

Published: unpublished

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Summary

The COSIMA Cookbook is an open computational learning module for analysing ocean and sea-ice model output in Jupyter notebooks. It has been developed by the Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA; <https://cosima.org.au>) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean-sea ice model configurations (e.g. [Kiss et al., 2020](#)). The repository combines introductory tutorials with worked analysis examples, all exposed through a browsable Sphinx documentation site ([Figure 1](#)) and backed by a citable archived release ([Constantinou et al., 2026](#)).

COSIMA Cookbook

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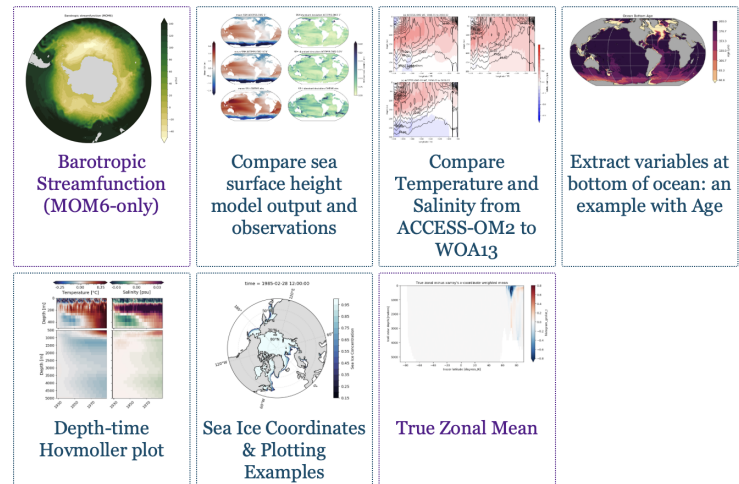
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Appetisers (easy)



latest

Figure 1: The COSIMA Cookbook documentation website, showing the browsable Sphinx-based landing page used to navigate tutorials and recipes. The live site is available at <https://cosima-recipes.readthedocs.io>.

Following the “Cookbook” concept, the website sections are deliberately named using a gastronomy theme. “Cooking Tutorials” refers to tutorials that teach generic, transferable techniques for handling ocean–sea ice model output. Then comes the main part of any cookbook, its recipes. Here, by “recipes” we mean self-contained notebooks showing how to perform concrete diagnostics and analyses on ocean and sea-ice datasets. First come the “Easy Recipes” that provide a good entry point after the tutorials; then the “Advanced Recipes” that are more elaborate and advanced analysis examples, and last the “Regional Specialties” contains recipes for regional model configurations (Barnes et al., 2024). Together these sections present the documentation as a browsable collection of lessons and recipes gathered into a single cookbook.

Pedagogy and instructional structure are central to the project. The idea is that these recipes provide the starting point for more elaborate analyses. “Cooking Tutorials” introduces generic skills such as loading model output, working with labelled arrays, plotting, and interacting with shared data catalogues. These tutorials lead into domain-focused recipes giving learners an incremental path from first contact with the data ecosystem to adaptation of full workflows for their own science questions.

Within this structure, the tutorials include examples of loading model output through Intake-based catalogues (Intake developers, 2026) and incorporating Cartopy (Elson et al., 2024) for map-based visualisation. The recipe collection also includes a Lagrangian particle-tracking workflow built with Parcels (Delandmeter & Sebille, 2019), as well as analyses that use xgcm (Abernathy et al., 2022) to handle variables defined on staggered finite-volume grids and xESMF (Zhuang et al., 2025) for regridding model output.

54 The Cookbook grew from a practical need inside the COSIMA community: the tools used
55 to analyse modern ocean model output are powerful, but the gap between package-level
56 documentation and a reproducible end-to-end workflow remains large. Users need to
57 understand not only Python and Jupyter (Kluyver et al., 2016), but also how to navigate
58 high-dimensional model output, shared high-performance computing environments, and
59 domain-specific analysis conventions. The Cookbook addresses that gap by packaging
60 reusable workflows in the same medium in which users actually work.

61 Statement of Need

62 Ocean and climate model analysis has a steep entry cost. New users must learn how
63 to find datasets, load them efficiently, interpret metadata, operate on multi-dimensional
64 arrays, and produce scientifically meaningful diagnostics. General-purpose libraries such
65 as xarray (Hoyer & Hamman, 2017) provide the foundations for these tasks, but they
66 do not by themselves show a learner how to translate a research question into a robust
67 analysis workflow for a specific model and computing environment.

68 The COSIMA Cookbook fills that need with a domain-specific, openly maintained collection
69 of computational lessons and examples. Its contribution is not a new analysis package;
70 rather, it is a reusable learning resource that makes existing scientific software, data
71 conventions, and model output more approachable. This aligns well with JOSE's emphasis
72 on open educational resources that enable computational learning through authentic
73 practice.

74 Several design decisions make the Cookbook particularly useful for adoption by other groups.
75 First, each notebook is intended to be self-contained and well-documented, so learners
76 can read, run, and modify a complete workflow without reconstructing missing context
77 from scattered notes. Second, the repository distinguishes tutorials from recipes. Tutorials
78 teach transferable skills, while recipes demonstrate how those skills combine in realistic
79 analysis tasks. Third, the project includes contributor guidance and notebook review
80 conventions that encourage new analyses to be written in a reusable and pedagogically
81 clear style. A caveat is that some recipes use high-resolution model output with datasets
82 of order terabytes, so the underlying data are not always practical to distribute or mirror
83 in full. However, beyond the initial data-loading step, which usually occurs early in each
84 recipe, the analysis workflows are generally applicable to most operational ocean-sea ice
85 model outputs that can be loaded with xarray (Hoyer & Hamman, 2017).

86 Thus, the structure is valuable beyond the immediate COSIMA context. Many research
87 communities maintain model output on shared infrastructure and face the same challenge:
88 turning expert tacit knowledge into examples that newcomers can adapt. The Cookbook
89 offers a model for how a scientific collaboration can capture that knowledge in version-
90 controlled notebooks, publish it as a living resource, and continuously improve it through
91 community contribution.

92 Educational Design and Experience of Use

93 The learning experience is organised as a progression. The documentation points new users
94 first to introductory lessons, including material on loading, slicing, and visualising model
95 output. Learners then move to more advanced tutorials and finally to recipe notebooks
96 that address concrete scientific questions. The categories of “easy” and “advanced” recipes
97 provide a lightweight pedagogical cue about expected complexity and scope, with “regional
98 specialties” specifically covering recipes for regional model configurations (Barnes et al.,
99 2024).

100 The intended mode of use is also explicit. The repository is designed around Jupyter-

101 based analysis on the Australian National Computational Infrastructure, with guidance
102 on running notebooks through ARE/JupyterLab sessions and on accessing shared data
103 holdings. This makes the Cookbook more than a static collection of examples: it is
104 operational documentation for a real analysis environment. At the same time, the
105 notebooks demonstrate broadly transferable patterns for working with labelled geophysical
106 data, so individual lessons can be reused or adapted outside that environment.

107 The project also supports community authorship. Contributors are encouraged to submit
108 new notebooks through pull requests, document their methods clearly, and generalise
109 workflows so that they remain useful to others. In that sense, the Cookbook functions
110 both as a learning module for end users and as a framework for teaching reproducible
111 scientific communication through notebook design and peer review.

112 Author order

113 The first two authors contributed equally; authors from the third position onward are
114 listed alphabetically by surname.

115 Acknowledgements

116 We are grateful to the COSIMA community for developing and maintaining the model
117 output, workflows, and pedagogical practices and also for the great community spirit that
118 defines COSIMA. We also acknowledge the National Computational Infrastructure, which
119 is supported by the Commonwealth government of Australia, and the ACCESS National
120 Research Infrastructure (ACCESS-NRI) ecosystem that make the operational use of these
121 notebooks possible. ACCESS-NRI's model ACCESS-OM2 is enabled by the Australian
122 Government's National Collaborative Research Infrastructure Strategy (a.k.a. NCRIS). We
123 acknowledge funding from the Australian Research Council under the Center of Excellence
124 for the Weather of the 21st Century CE230100012, the Linkage Infrastructure, Equipment
125 and Facilities LP160100073 and LP200100406, and the Discovery Project DP240101274.

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